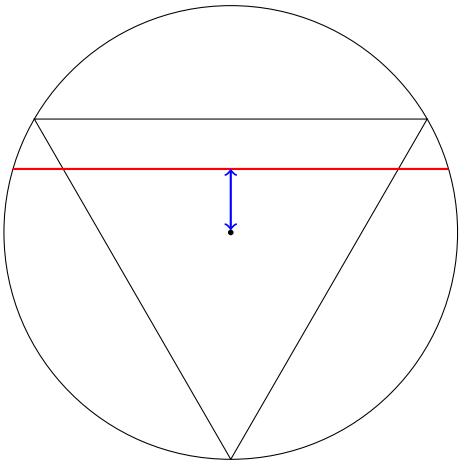
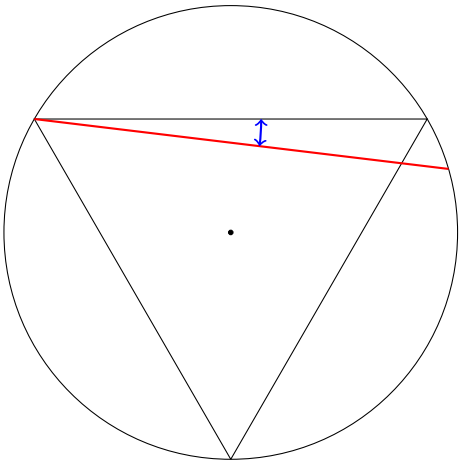
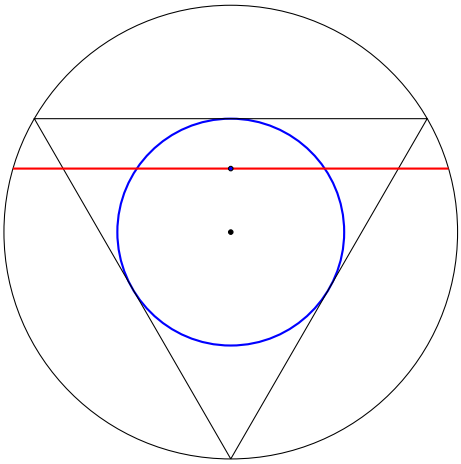
Distance from center: $\frac{1}{2}$.



Angle: $\frac{1}{3}$.



From a random point: $\frac{1}{4}$.